

Conservative divergence-conforming state transfer from adaptive potential-flow BIE to embedded VOF: overturning waves over variable bathymetry

Saveliy Baturin

Abstract

A structure-preserving one-way state transfer is developed from an adaptive fully nonlinear Euler boundary-integral equation (BIE) solver to an embedded two-phase volume-of-fluid (VOF) receiver. The contribution is not BIE–VOF hybridisation itself, but a handoff that matches the receiver’s discrete invariants. Integrated panel circulations preserve the adaptive BIE state; shared face-averaged BIE fluxes are fitted by a vertex streamfunction so that the MAC face field is divergence-free by construction; receiver momentum is included as a quadrature constraint; and the full-cell liquid aperture $q = fc_s$ is transported with a shared embedded-boundary flux. The BIE also uses a nonlocal-gap trigger for oversampled close evaluation. After the single transfer, only the VOF solver advances the solution.

Verification separates operator accuracy, conservation and interface geometry. On a manufactured folded boundary, oversampling reduces the relative Dirichlet-to-Neumann error from 6.36 to 6.91×10^{-2} at 64 panels despite similarly small linear residuals. A solitary-wave translation has shape and speed errors 1.82×10^{-5} and 8.40×10^{-5} . Manufactured aperture transport is first-order on the three finest grids and preserves liquid volume and $0 \leq q \leq c_s$ to roundoff. Across the three production receiver grids, the constrained handoff has at most 1.56% face-flux error, 2.78% surface-velocity error and 7.09×10^{-9} relative receiver-momentum error.

An overturning-wave state over variable bathymetry is continued to nondimensional time 6 on three nested receiver grids. Maximum relative liquid-volume drift is 8.56×10^{-8} , 6.43×10^{-8} and 5.67×10^{-9} on levels 8–10, with no aperture-bound violation. Level 10 produces a transient disconnected gas component at $t = 1.96$, but the event is absent at the two coarser levels and its detected onset varies by 0.06 under conservative raster coarsening. At $t = 6$, the level-9/10 upper-profile and crest differences are 8.18% and 6.25%. The method therefore achieves stable conservative continuation and exposes a resolution-dependent topology transition, but the present calculation does not establish grid-converged wave impact or air entrainment. The verified result is therefore a conservative, divergence-conforming model transition rather than a converged prediction of breaking topology.

Keywords: conservative state transfer; divergence-conforming transfer; boundary integral method; volume-of-fluid method; embedded boundaries; adaptive mesh refinement; free-surface topology change

1 Introduction

Boundary-integral descriptions of inviscid water waves resolve vertical and overturning free surfaces without a height-function restriction [1, 6, 16]. They remain efficient while the liquid is irrotational and the interface is a single non-self-intersecting curve, but cease to describe entrained air, vorticity production and topology change after impact. Fully nonlinear BEM calculations have reproduced solitary-wave shoaling and classified breaking regimes over beaches and obstacles [8], while two-phase VOF methods naturally continue through interface reconnection.

Hybridisation itself is well established. Lachaume et al. initialised a Navier–Stokes/VOF breaking-wave calculation from a potential-flow BEM and compared solitary-wave evolution

Table 1: Direct predecessors and the scope of the present transition. “Not posed” means that the cited method did not formulate the listed property as an exact constraint of the state-transfer operator; it does not imply that the subsequent solver was non-conservative.

Study	Architecture	Field receiver	Information exchanged	Transfer properties emphasised
Lachaume et al. [15]	one-way initialisation	Navier–Stokes/VOF	interface and BEM flow state	breaking-wave continuation and experimental comparison
Colicchio et al. [5]	simultaneous domain decomposition	Navier–Stokes/level set	interface data across overlapping subdomains	robust large deformation and fragmentation
Kim et al. [14]	two-way domain coupling	SMAC/VOF	bidirectional interface-boundary data	long random-wave propagation
Khait and Ma [13]	coupled potential/viscous model	two-phase Navier–Stokes/VOF	potential-wave forcing and viscous response	breaking-energy and non-potential effects
Present work	one-time state transfer	embedded MAC/VOF with AMR	liquid aperture, shared face fluxes and total momentum	exact liquid measure and MAC divergence; constrained momentum and circulation

with experiments [15]. Colicchio, Greco and Faltinsen coupled BEM and Navier–Stokes/level-set subdomains for deforming and fragmented interfaces [5]. Kim, Yamashiro and Yoshida developed a two-way BEM–SMAC/VOF decomposition for long random-wave calculations [14]. More recently, Khait and Ma combined fully nonlinear quasi-potential and two-phase Navier–Stokes models for energetic breaking waves [13], while Kanehira et al. embedded a GPU-SPH region in a three-dimensional fully nonlinear potential-flow calculation [12]. These studies establish the value of hybrid models; they also make clear that coupling as such is not a novelty claim.

Table 1 separates the architectures from the narrower question addressed here. Earlier work focuses on domain exchange, initialisation or physical validation. The present work instead asks whether a late interface-and-velocity state can be handed once to a non-conforming embedded receiver while satisfying that receiver’s finite-volume constraints.

The contribution is expressed as four testable statements. (i) Conservative redistribution preserves every integrated panel circulation and hence the total boundary circulation to roundoff. (ii) A shared-face streamfunction curl has identically zero Cartesian finite-volume divergence before receiver sampling. (iii) receiver-grid liquid momentum is imposed within the sparse reconstruction rather than repaired by an unconstrained posterior shift. (iv) Advecting $q = fc_s$ with one geometric flux per shared face makes the global physical liquid measure telescope and preserves $0 \leq q \leq c_s$. The calculations below test these statements and then apply the transition to an overturning breaker on three receiver grids.

2 Mathematical formulation

2.1 Pre-contact free-surface flow

Let $\Omega(t) \subset \mathbb{R}^2$ be the liquid domain. With velocity $\mathbf{u} = \nabla\phi$, the potential satisfies

$$\nabla^2\phi = 0 \quad \text{in } \Omega(t), \quad \partial_n\phi = 0 \quad \text{on the bed } \Gamma_b. \quad (1)$$

The periodic free surface is $\mathbf{X}(\alpha, t) = (x(\alpha, t), z(\alpha, t))$ and carries the trace $\psi = \phi|_{\Gamma_f}$. With normal velocity $V_n = \partial_n\phi$, tangential fluid velocity $V_t = \partial_s\psi$ and freely chosen marker velocity W_t , the surface equations are

$$\mathbf{X}_t = V_n \mathbf{n} + W_t \mathbf{t}, \quad (2)$$

$$\psi_t = \frac{1}{2}V_n^2 - \frac{1}{2}V_t^2 + W_t V_t - gz. \quad (3)$$

For prescribed ψ_f and zero bed flux, Green's identity gives the mixed system

$$[S_{:f} \quad -(\frac{1}{2}I + K)_{:b}] \begin{bmatrix} q_f \\ \phi_b \end{bmatrix} = (\frac{1}{2}I + K)_{:f} \psi_f, \quad (4)$$

where $q_f = \partial_n \phi|_{\Gamma_f}$ and S and K are the periodic single- and double-layer operators. Their Green function is

$$G_L(\Delta x, \Delta z) = -\frac{1}{4\pi} \log \left[2 \left(\cosh \frac{2\pi \Delta z}{L} - \cos \frac{2\pi \Delta x}{L} \right) \right]. \quad (5)$$

Kress product weights integrate the logarithmic self interaction. The incident-wave normal velocity is projected onto the actual bathymetry by subtracting its mean boundary flux and solving the compatible Neumann problem in a zero-mean gauge.

The ordinary self rule becomes under-resolved when two distant arclength segments approach. Define

$$d_{\text{nl}} = \min_{d_s(i,j) > 8\Delta s} \|\mathbf{X}_i - \mathbf{X}_j\|, \quad \overline{\Delta s} = \frac{2\pi}{N} |\overline{\mathbf{X}_\alpha}|. \quad (6)$$

When $d_{\text{nl}}/\overline{\Delta s} < 1.5$, the complete free-surface self block is reevaluated after trigonometric interpolation to pN points, with

$$p = \text{clip}_{[2,16]} \lceil 4\overline{\Delta s}/d_{\text{nl}} \rceil. \quad (7)$$

The density and geometry are oversampled together so that the Kress splitting remains consistent.

2.2 Adaptive circulation representation

The surface potential trace is equivalently represented by sheet strength $\gamma = -\partial_s \psi$. For a panel j the integrated circulation

$$\Gamma_j = - \int_{s_j}^{s_{j+1}} \gamma \, ds = \psi_{j+1} - \psi_j \quad (8)$$

is used as the conservative carrier during marker redistribution. New panel endpoints are chosen from a smoothed monitor depending on curvature, sheet strength and inverse nonlocal gap; the cumulative circulation is interpolated monotonically and then differenced on the new partition. This preserves total circulation to roundoff. These boundary circulations represent the irrotational potential trace; they are not bulk material vortices.

2.3 Conservative embedded liquid aperture

For a Cartesian receiver cell K_i , define the full-cell physical liquid aperture

$$q_i = \frac{|K_i \cap \Omega_\ell \cap \Omega_f|}{|K_i|}, \quad 0 \leq q_i \leq c_{s,i}, \quad (9)$$

where c_s is the open-fluid volume fraction and $f = q/c_s$ is conditional on the open part of the cell. This full-cell measure is consistent with solid-aware VOF/embedded-boundary formulations and conservative contact-line treatments [4, 10, 11, 21]. One oriented swept volume is assigned to every shared face,

$$F_e = |S_e(\Delta t) \cap \Omega_\ell \cap \Omega_f|, \quad q_i^{n+1} = q_i^n - |K_i|^{-1} \sum_{e \subset \partial K_i} \text{sgn}(i, e) F_e. \quad (10)$$

Because the same F_e enters adjacent cells with opposite signs, global liquid volume telescopes exactly. A pure donor satisfies $F_{\ell,e} = 0$ for $f_d = 0$ and $F_{\ell,e} = F_{f,e}$ for $f_d = 1$; only genuinely mixed cut cells require a capacity restriction. The restriction is applied to mixed cells and their face-connected one-cell halo,

$$\Delta t \leq 0.45 \min_{i \in \mathcal{I}} \frac{c_{s,i} \Delta_i}{\max_{e \subset \partial K_i} |u_e f_{s,e}|}. \quad (11)$$

2.4 Receiving two-phase model and discretisation

After the handoff the BIE is no longer advanced and no feedback is returned to it. In the open part of the embedded domain, the receiver solves the dimensionless variable-density incompressible equations

$$\nabla \cdot \mathbf{u} = 0, \quad (12)$$

$$\partial_t(\rho \mathbf{u}) + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \nabla \cdot (2\mu \mathbf{D}) + \rho \mathbf{g} + \sigma \kappa \delta_\Gamma \mathbf{n}, \quad (13)$$

$$\partial_t f + \nabla \cdot (f \mathbf{u}) = 0, \quad \mathbf{D} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top). \quad (14)$$

Here f is conditional on open fluid, while equation (10) advects the physical aperture $q = f c_s$. Arithmetic mixture laws are used,

$$\rho(f) = f \rho_\ell + (1 - f) \rho_g, \quad \mu(f) = f \mu_\ell + (1 - f) \mu_g. \quad (15)$$

Length is scaled by the offshore depth h_0 , velocity by $\sqrt{g h_0}$ and time by $\sqrt{h_0/g}$. Thus $Fr = 1$, and the reported runs use

$$\rho_\ell = 1, \quad \rho_g/\rho_\ell = 1/850, \quad Re = 40000, \quad \mu_g/\mu_\ell = \frac{17.4 \times 10^{-6}}{8.9 \times 10^{-4}}, \quad Bo = We = 1000. \quad (16)$$

Surface tension is therefore present with $\sigma = 1/Bo$; gravity is $\mathbf{g} = (0, -1)$. The implementation uses a balanced VOF surface-force form and reduced pressure for gravity [18]. The interface and mixture properties follow the geometrical VOF formulation [9, 18].

The receiver is the cell-centred adaptive projection method of Popinet [17], with a Bell–Colella–Glaz predictor [2], an implicit viscous solve and a multilevel pressure Poisson solve. The two phase momenta $f \rho_\ell \mathbf{u}$ and $(1 - f) \rho_g \mathbf{u}$ are carried with the same geometrical phase flux and recombined after advection. The pressure projection tolerance is 10^{-8} . The accepted step is limited by the ordinary advective CFL, the explicit capillary condition and equation (11).

The horizontal direction is periodic. The embedded bathymetry is impermeable and free slip ($u_n = 0$, $\partial_n u_t = 0$); the distant top of the $64h_0$ -high computational box uses the corresponding homogeneous symmetry condition. Pressure is not transferred: it is initialised to zero and the first receiver projection constructs the dynamic pressure compatible with the mapped velocity. After every accepted step, wavelet adaptation uses absolute tolerances 2×10^{-4} for q , 2×10^{-3} for each velocity component and 8×10^{-3} for c_s . Maximum/minimum level pairs are (8, 6), (9, 7) and (10, 8). The conservative q restriction is applied before momentum prolongation, retained face fluxes are reprojected after a mesh change, and the associated momentum and energy changes are recorded.

2.5 Divergence-conforming state transfer

The target liquid momentum is evaluated from boundary data,

$$\mathbf{P}^\star = \rho \int_\Omega \nabla \phi \, dA = \rho \oint_{\partial\Omega} \phi \mathbf{n} \, ds. \quad (17)$$

Two-point Gauss quadrature supplies the BIE velocity averaged over every receiver face. A periodic vertex streamfunction χ then defines

$$u_{i-1/2,k} = \frac{\chi_{i-1/2,k+1/2} - \chi_{i-1/2,k-1/2}}{\Delta z}, \quad w_{i,k-1/2} = -\frac{\chi_{i+1/2,k-1/2} - \chi_{i-1/2,k-1/2}}{\Delta x}. \quad (18)$$

The standard face-difference divergence of this field vanishes identically. Compatibility projections and divergence-preserving transfer between non-conforming or multilevel grids provide the relevant numerical context [3, 7, 20]. The sparse fit minimises

$$\|W_f(C_h \chi - \bar{\mathbf{u}})\|_2^2 + \lambda_s \|S\chi - \mathbf{u}_s\|_2^2 + \lambda_b \|B\chi\|_2^2 + \lambda_P \|M\chi - \mathbf{P}^\star/\rho\|_2^2 + \lambda_e \|E\chi\|_2^2 + \lambda_h \|L_h \chi\|_2^2. \quad (19)$$

Table 2: Dimensionless per-row squared-residual multipliers in the production state-transfer fit.

Term	Weight	Purpose
$W_f(C_h\chi - \bar{\mathbf{u}})$	1	trusted shared-face BIE fluxes
$S\chi - \mathbf{u}_s$	$\lambda_s = 32$	free-surface velocity trace
$B\chi$	$\lambda_b = 10^4$	bed streamfunction
$M\chi - \mathbf{P}^*/\rho$	$\lambda_P = 10^8$	receiver-grid momentum
$E\chi$	$\lambda_e = 5 \times 10^{-3}$	exterior kinetic energy
$L_h\chi$	$\sqrt{\lambda_h} = 0.20 \min(\Delta x, \Delta z)$	harmonic regularisation

Here $B\chi = 0$ makes the bed a streamline, M imposes momentum, E suppresses exterior kinetic energy and L_h regularises unconstrained modes. After AMR, the face flux transported by the mesh is retained and a receiver-grid Helmholtz correction restores compatibility; no global velocity shift is applied. Table 2 gives the production weights. The face term has unit weight; the Laplacian row is scaled before squaring, so the last entry is reported as $\sqrt{\lambda_h}$. All coordinates and fields in this fit are nondimensional. The reported λ values are per-row squared-residual multipliers: each block is premultiplied by $\sqrt{\lambda}$, with no division by its number of rows. Consequently their absolute magnitudes are not cross-block physical importance factors, and the aggregate block contributions change with row count under refinement. A right diagonal column equilibration scales every unknown by the inverse Euclidean norm of its assembled column for the iterative solve and is undone before χ is returned; it does not alter the row weights. One set of weights was fixed before the three-grid runs and used without level-by-level retuning, so the reported grid results test that fixed discrete choice rather than a resolution-invariant hyperparameter normalisation.

3 Verification

The tests are grouped by the property they measure. Analytic and manufactured solutions test the BIE and aperture operators; algebraic residuals test the handoff constraints; long receiver runs test conservation and boundedness. Interface contact is assessed separately in Section 4.

For a flat surface Fourier mode, the Dirichlet-to-Neumann eigenvalue and the discrete velocity agree to machine precision. A fully nonlinear solitary wave propagated over one reference interval has relative shape error 1.82×10^{-5} and speed error 8.40×10^{-5} . On a manufactured folded interface, the unoversampled 64-panel discretisation has a relative DNO error of 6.36 even though its linear residual is near 10^{-13} ; order-16 oversampling reduces the error to 6.91×10^{-2} (figure 1). Repeating the factor sweep at 128 panels reduces the unoversampled error from 2.74 to 3.56×10^{-2} at factors 8–16. The plateau under further oversampling identifies the remaining error as boundary-discretisation rather than linear-solver error. This two-grid, one-gap study verifies the trigger mechanism but is not a complete $(N, d_{\text{nl}}/h, p)$ convergence map.

For sinusoidal-interface translation, the relative $L^1(q)$ errors on levels 6–8 are 0.01378, 0.007689 and 0.003538, corresponding to observed orders 0.84 and 1.12. Fixed, frozen-AMR and every-step-AMR volume drift remains below 2.0×10^{-13} . An inclined-wall test gives errors 0.02325, 0.01247 and 0.006154 on levels 7–9 (orders 0.90 and 1.02) while preserving volume and bounds to roundoff. Restricting the cut-cell timestep to the interface band requires 211 steps at level 7, compared with 758 when every cut cell is included.

The q construction is also compared directly with independently updating the product fc_s in a one-step inclined-wall kinematic reference (table 3). The product update both changes the physical liquid volume and incurs a transport error; the shared- q update telescopes to roundoff on all three grids. This is an operator test, not a substitute for the coupled receiver experiment below.

Independent cell-centred samples of the late BIE velocity have a 15.92% best-fit mismatch when represented in the range of the receiver curl. This cell-to-curl-space quantity is not the

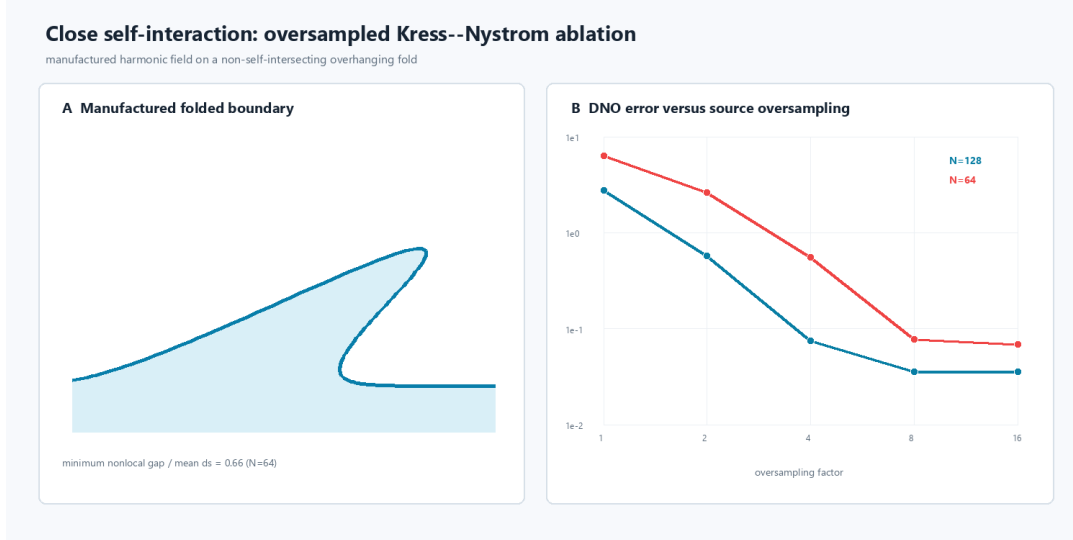


Figure 1: Close-interaction quadrature on a manufactured folded interface. The boundary-integral residual alone does not diagnose the unresolved parameter-distant branch; geometric oversampling reduces the DNO error.

Table 3: Ablation of the physical liquid measure in the kinematic reference. $\delta V/V$ is the signed relative volume change and the final column is the largest violation of $0 \leq q \leq c_s$ for the shared- q scheme.

N	product L^1	product $\delta V/V$	shared- q L^1	shared- q $\delta V/V$	bound error
32	1.85×10^{-3}	7.76×10^{-5}	6.31×10^{-15}	0	3.33×10^{-16}
64	4.31×10^{-4}	-9.45×10^{-5}	1.59×10^{-14}	0	3.57×10^{-16}
128	9.09×10^{-5}	-3.59×10^{-6}	3.63×10^{-14}	0	7.81×10^{-16}

face-flux error used for the constrained rows of table 4, so the corresponding table entry is left blank rather than compared numerically. Shared face averages followed by equation (19) give a face-flux error of 1.6835% and a surface-velocity error of 3.1229%. Relative momentum error is 1.99×10^{-9} and the finite-volume divergence RMS is 1.07×10^{-14} . GPU sparse LSQR takes 12.76 s versus 69.54 s for the CPU solve and changes the trusted-face field by only 4.58×10^{-8} relative (figure 2). This acceleration applies to the sparse handoff fit; the two-phase evolution reported below is CPU-serial.

Table 4 resolves the transfer into progressively constrained variants on the same pre-contact BIE snapshot. A curl fit is already divergence conforming; surface, exterior and harmonic terms trade local data fit for a controlled continuation outside the trusted liquid region. Adding harmonic regularisation raises the face and surface errors, but keeps momentum below 2×10^{-9} and eliminates unconstrained high-frequency modes. These rows are a bounded one-state sensitivity study, not an optimisation over weights.

4 Three-grid overturning-wave experiment

4.1 Configuration and event definition

The source is an active $N = 576$ BIE state over a periodic 1:30 beach and return slope, transferred at nondimensional BIE time 20.303125. Conservative raster overlap changes volume by 2.45×10^{-13} relative to the source; reconciliation with the analytic embedded bed changes 0.0868%. Receiver maximum levels 8, 9 and 10 use minimum levels 6, 7 and 8, respectively, and adapt after every accepted physical step. VOF and vorticity fields are sampled every 0.02 through continuation

Table 4: State-transfer ablation at BIE time 20.29375. Errors are relative; dashes denote a quantity not imposed or not meaningful for that representation.

Representation	face error	surface error	momentum error	divergence RMS
independent cell samples	–	–	–	8.37×10^{-1}
raw shared-face averages	–	–	–	1.24×10^{-2}
streamfunction curl, face only	0.1307%	–	–	1.05×10^{-14}
+ bed and momentum	0.1307%	–	6.32×10^{-14}	1.05×10^{-14}
+ surface and exterior	0.1310%	0.700%	1.26×10^{-9}	1.05×10^{-14}
full fit with harmonic term	1.6835%	3.1229%	1.99×10^{-9}	1.08×10^{-14}

Table 5: Three-grid receiver continuation through $t = 6$. $A_{\max}$ is the largest exterior-disconnected, bed-detached gas area before the one-cell area filter.

Level	$\max \delta V/V $	$\max D_h \mathbf{u} _\infty$	$\max \delta P/P $	$A_{\max}/\Delta^2$	resolved onset
8	8.56×10^{-8}	3.59×10^{-10}	8.24×10^{-5}	0	none
9	6.43×10^{-8}	2.32×10^{-10}	4.04×10^{-5}	0.498	none
10	5.67×10^{-9}	3.86×10^{-9}	6.40×10^{-5}	10.57	1.96

time 6.

The receiver-quadrature fits at levels 8–10 have face-flux errors 1.5496%, 1.5505% and 1.5510%, surface-velocity errors 2.7794%, 2.7791% and 2.7790%, and relative receiver-momentum errors between 6.37×10^{-9} and 7.09×10^{-9} . Their weighted velocity changes from the unconstrained receiver samples are 1.05%, 1.19% and 2.01%, while the corresponding kinetic-energy changes are 0.279%, 0.252% and 0.320%. These results quantify the price of imposing momentum on each actual adaptive receiver quadrature rather than only on the uniform export grid.

A preliminary level-9 two-time screen compared BIE states at 20.29375 and 20.303125 through receiver time 0.30. It used the pre-production receiver in which f and c_s were updated independently and multiplied, although the post-AMR face flux was preserved; it did not use the shared- q transport of the three-grid experiment. The maximum profile and crest differences in that legacy screen are 0.944% and 1.22%, while the later run’s maximum volume drift is 1.52×10^{-3} . By contrast, the production shared- q level-9 run at the same BIE time has maximum drift 6.43×10^{-8} through $t = 6$. The values therefore describe different discrete transport operators, not two estimates of one run. The legacy screen is retained as negative provenance only; production-scheme sensitivity to handoff time remains untested.

A gas component is classified as a resolved closure if it is disconnected from the exterior, does not touch a one-finest-cell bed guard, has area at least Δ^2 , and persists for three output frames. Four-neighbour components are joined across the horizontal periodic seam. Diagnostic coarsening uses exact block liquid-area fractions at scale factors two and three and thresholds 0.25, 0.5 and 0.75. The accepted onset-time variation is at most 0.04.

4.2 Conservation and topology

All three continuations remain bounded to $t = 6$. Table 5 summarises the invariant and compatibility histories. The aperture bounds hold to roundoff. The largest post-AMR divergence and relative momentum and kinetic-energy corrections remain below 3.86×10^{-9} , 8.24×10^{-5} and 9.31×10^{-6} , respectively.

Level 8 has no disconnected gas component. Level 9 develops a late component whose maximum area is only $0.498\Delta_9^2$. Level 10 develops a component larger than ten finest-cell areas near $t = 1.96$, but it has disappeared by $t = 2.08$. Across the seven conservative raster probes its onset ranges from 1.90 to 1.96, exceeding the 0.04 tolerance, as summarised with the invariant histories in figure 3. A later group of cavities at $t = 5.10$ is likewise absent on the coarser grids, as the matched VOF masks in figure 4 show directly. Thus the algebraic conservation properties converge more strongly than the observed topology.

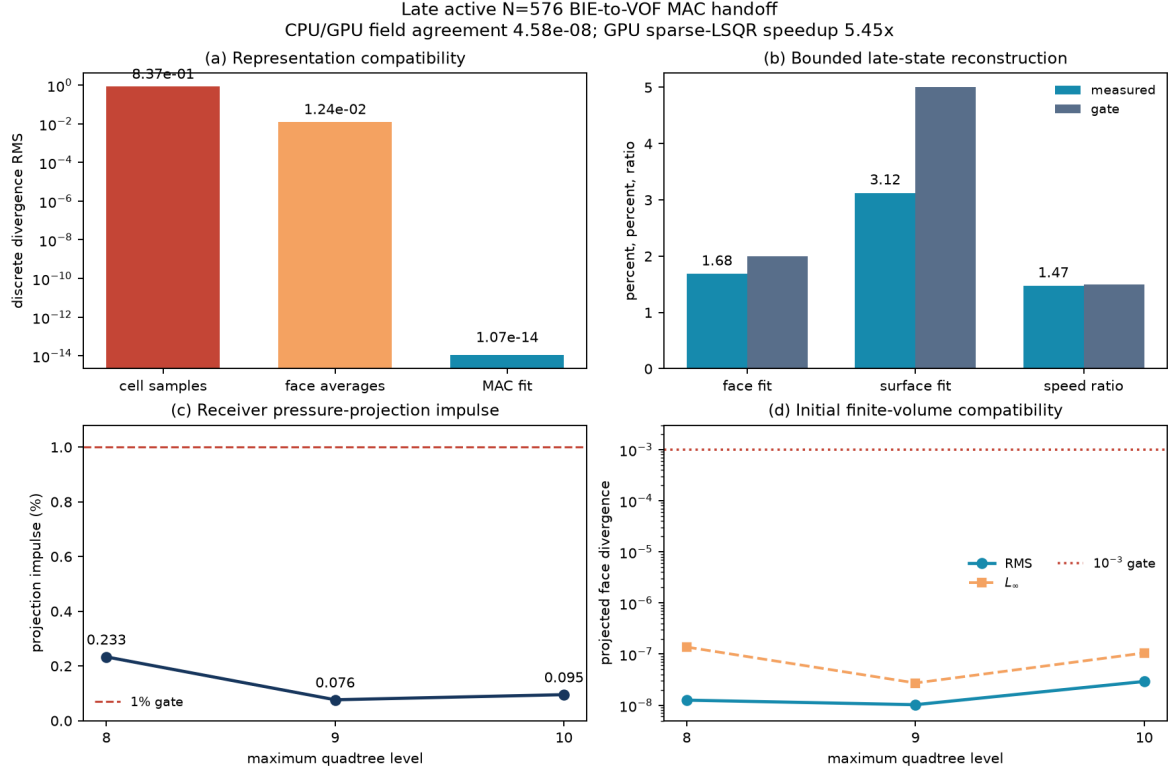


Figure 2: Compatible controlled transfer at BIE time 20.29375, used for the ablation rather than the production three-grid handoff. Face averaging reduces the initial representation mismatch and the streamfunction curl enforces finite-volume divergence compatibility. Receiver mass, momentum and post-projection divergence are shown for levels 8–10.

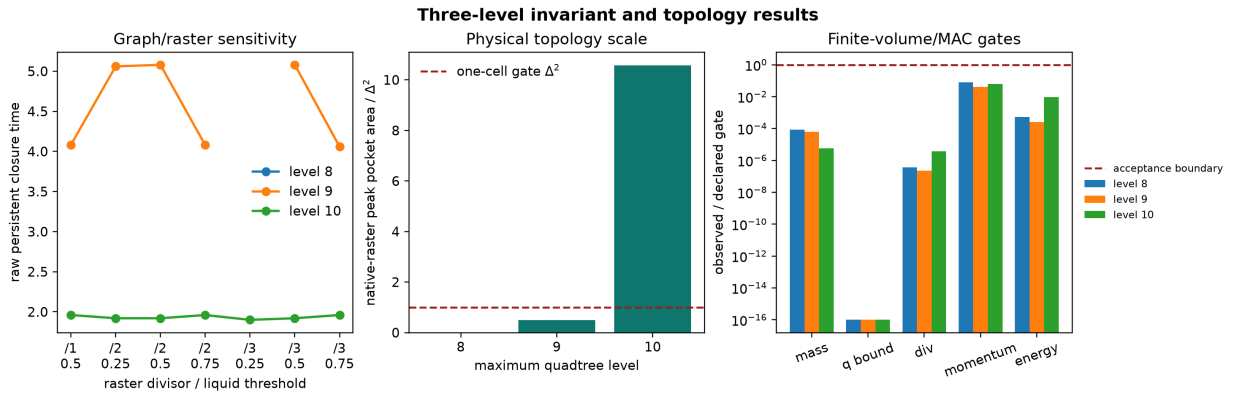


Figure 3: Raster sensitivity, gas-component scale and normalized conservation and MAC compatibility measures. The level-10 component is physically larger than one finest cell, but its time and existence are not reproduced across the receiver grids.

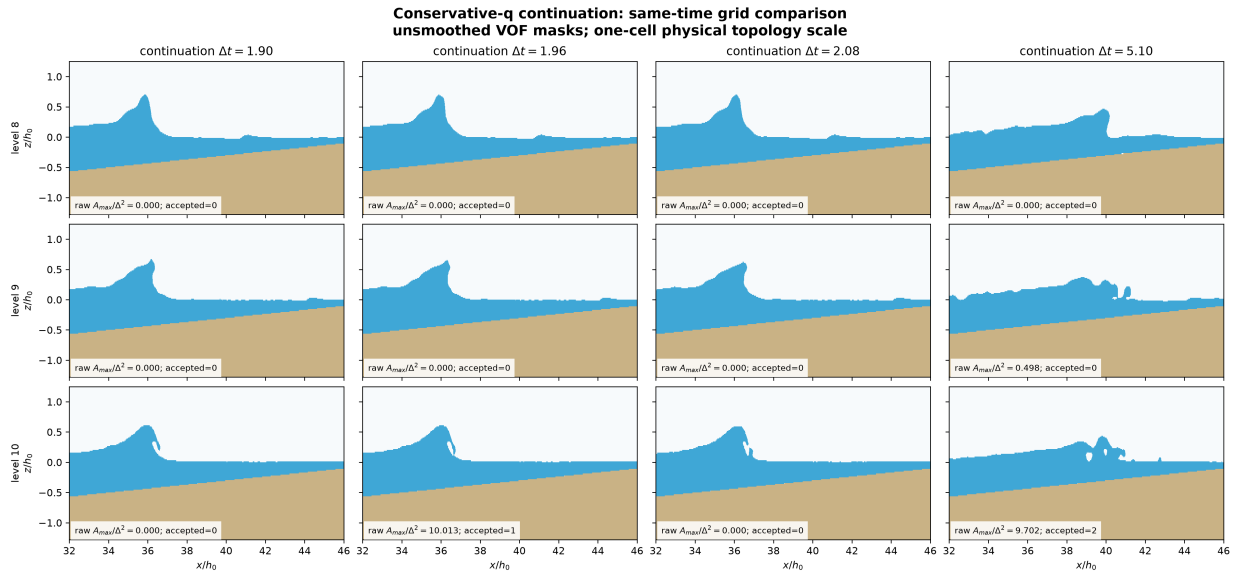


Figure 4: Unsmoothed VOF masks at matched continuation times. The transient level-10 cavity at $t = 1.96$ and the later cavities at $t = 5.10$ are not same-time features of levels 8 and 9. Labels report the largest raw component in units of the local finest-cell area and the number passing the physical area filter.

4.3 Geometric refinement

Upper-surface profiles are extracted at identical physical times without interpolation between frames. Relative RMS differences are normalised by the initial level-10 amplitude. The maximum L8/L9 and L9/L10 profile differences over $0 \leq t \leq 6$ are 11.39% and 8.26%; at $t = 6$ they are 8.12% and 8.18%. The final L9/L10 crest difference is 6.25%, with a maximum of 12.5%. Although the instantaneous Richardson indicator is positive for 80.4% of the last time unit, the fine-pair profile and crest differences both exceed the 5% tolerance (figure 5).

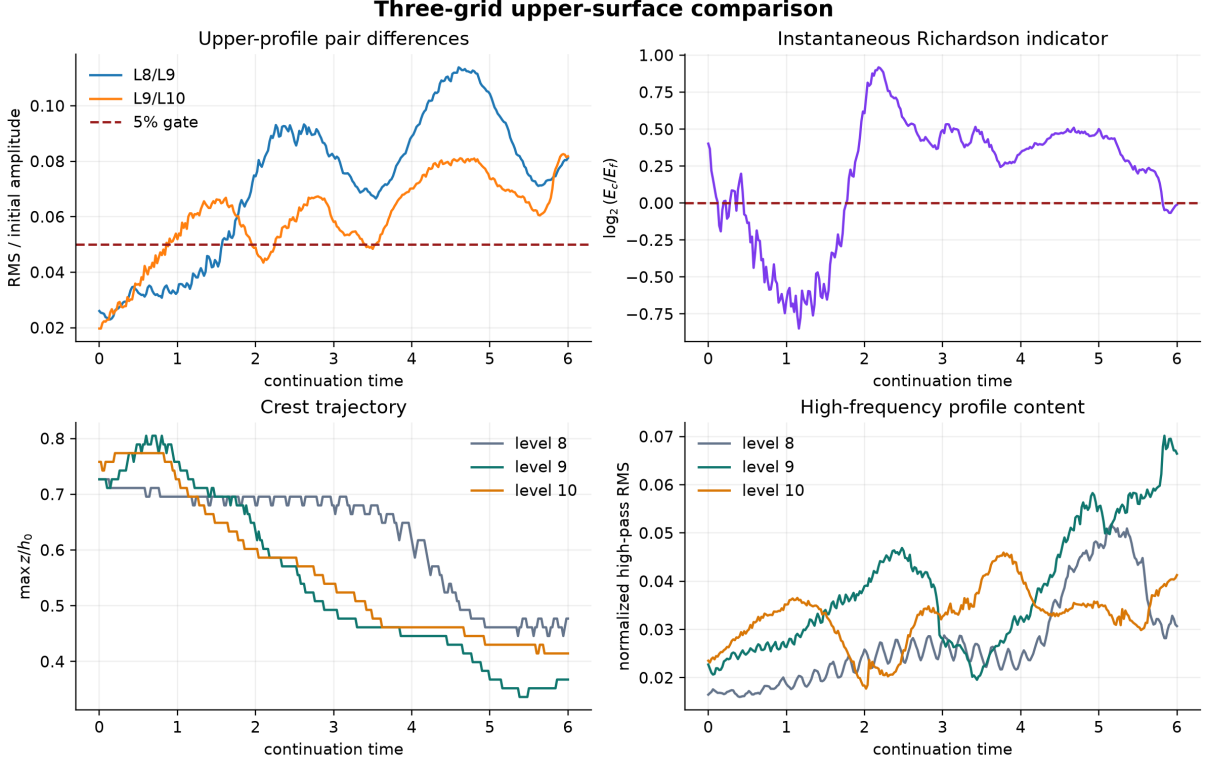


Figure 5: Three-grid upper-surface comparison. Pairwise profile errors remain above the 5% tolerance for much of the continuation, and the final fine-pair profile and crest differences are 8.18% and 6.25%.

The mismatch is not captured by smooth-profile measures alone. On level 9, adapting every second or fourth step changes the upper-profile RMS by at most 3.55% but shifts the first raw subcell component from $t = 4.08$ to about 5.1. Topology is therefore more sensitive to mesh evolution than the bulk profile.

The level-10 vorticity field contains a concentrated signed layer around the curling crest and beneath the emerging cavities (figure 6), consistent with the localisation reported in small-viscosity plunging-wave calculations [19]. The plotted markers sample the Eulerian vorticity quadrature $\Gamma_{ij} \simeq \omega_{ij} \Delta A$; they are not Lagrangian particles. Quantitative post-contact circulation requires a separate refinement study because potential-flow sheet labels cannot be continued through reconnection.

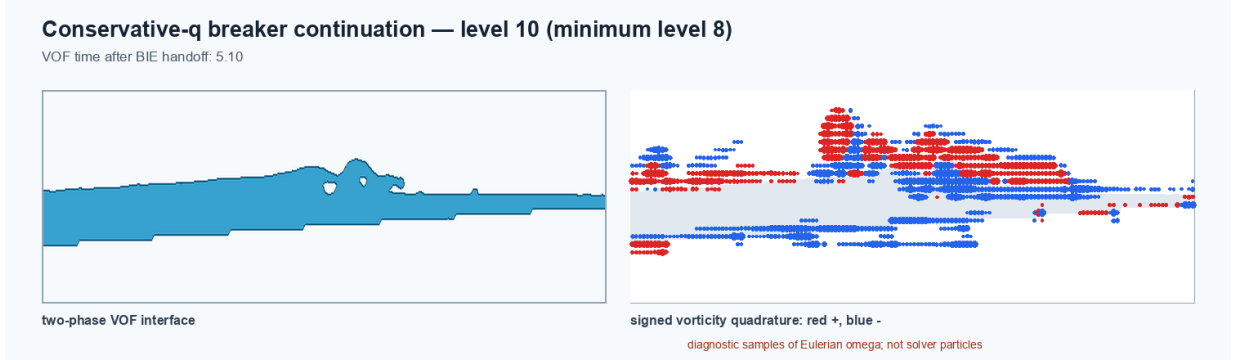


Figure 6: Level-10 state at $t = 5.10$. Left: two-phase VOF interface. Right: signed samples of the resolved Eulerian vorticity (red positive, blue negative).

5 Discussion

The experiment separates stability from physical convergence. Transporting q with shared embedded fluxes removes the previous long-time mass loss, and the receiver-grid correction retains divergence and momentum compatibility after AMR. These properties remain stable through 301 output times on all three grids. They do not, however, determine the topology of the free surface: the first resolved gas component appears only on level 10 and is transient, while the upper-surface geometry remains outside the selected fine-grid tolerance.

The level-10 cavity may be the first resolved representation of jet closure, or it may be a resolution-dependent branch selected by the coupled AMR and interface reconstruction. The current data cannot distinguish these possibilities. A decisive impact calculation requires at least one finer receiver grid, time-step refinement, several earlier handoff times and an aligned experimental or high-resolution two-phase reference. Event time, jet-tip position, enclosed-gas area and pressure impulse should converge together. Air compressibility and three-dimensionality may also become important near closure.

The contribution is therefore the conservative and divergence-conforming state transfer, together with a three-level observation that topology converges more slowly than invariants. The available ablations isolate cell-centred versus shared-face transfer, receiver curl reconstruction, momentum and surface constraints, harmonic regularisation, product versus shared- q transport, plus a negative two-time screen of the legacy product receiver. They do not yet constitute a full multi-parameter sensitivity analysis: production-scheme handoff timing, wider weight ranges, several nonlocal gaps, time-step refinement and a second physical scenario remain open. A matched end-to-end accuracy–cost comparison with a contemporary BEM–VOF or potential-flow/particle hybrid remains necessary to establish an efficiency advantage. The present GPU timing concerns only the sparse handoff reconstruction; the adaptive two-phase receiver runs on the CPU with OpenMP. In a fixed $t \leq 0.04$ repeatability screen, four threads reduced the level-10 wall time from 57.95 to 19.82 s (2.92-fold) relative to one thread. The final thresholded VOF mask was identical and the relative mass difference was 4.56×10^{-15} , while the final kinetic energy differed by 2.35×10^{-5} because OpenMP reductions are not bitwise deterministic. On level 9, eight threads were slower than four. These short runs select a performance configuration; they are not grid-convergence data.

6 Conclusions

An adaptive Euler–BIE state can be transferred to an embedded two-phase VOF receiver without sacrificing the receiver’s discrete divergence structure. The essential choices are to transfer shared face averages, construct the MAC field as a streamfunction curl, impose momentum in the receiver

quadrature and transport the full-cell liquid aperture $q = fc_s$ with a shared embedded-boundary flux.

Manufactured tests show first-order aperture convergence with roundoff mass and bound preservation. The L8–L10 receiver continuation remains bounded through $t = 6$, with maximum relative liquid-volume drift below 8.57×10^{-8} and small post-AMR compatibility corrections. The level-10 trajectory contains a resolved but transient gas component and strong crest vorticity. Its onset and morphology are not reproduced on the coarser grids, and the final L9/L10 profile and crest differences exceed 5%.

The present result supports a conservative, divergence-conforming one-way model transition and identifies a resolution-dependent topology transition. It does not yet support a grid-converged prediction of plunging impact, trapped-air volume or impact impulse.

Data and code availability

Source code, solver inputs, machine-readable diagnostics and figure scripts are available at https://github.com/Safreliy/wave_research under tag v0.29.0-openmp-visualization. The compact repository contains the exact L8–L10 receiver inputs and diagnostic logs. The 1,509 raw VOF/vorticity frames (approximately 503 MB) are not yet deposited in a DOI-bearing archive; they are required only to recompute connected components and profiles from pixels, and must be archived before journal submission. Research code is licensed under the MIT License, and the original manuscript, figures and derived data under CC BY 4.0; third-party software and data retain their own licences. The two-phase calculations use Basilisk patch ad05f362 in the immutable container image identified in the repository.

References

- [1] G. R. Baker, D. I. Meiron, and S. A. Orszag. Generalized vortex methods for free-surface flow problems. *Journal of Fluid Mechanics*, 123:477–501, 1982. doi: 10.1017/S0022112082003164.
- [2] John B. Bell, Phillip Colella, and Harland M. Glaz. A second-order projection method for the incompressible Navier–Stokes equations. *Journal of Computational Physics*, 85(2):257–283, 1989. doi: 10.1016/0021-9991(89)90151-4.
- [3] Antonio Cervone, Sandro Manservigi, and Ruben Scardovelli. An optimal constrained approach for divergence-free velocity interpolation and multilevel VOF method. *Computers & Fluids*, 47(1):101–114, 2011. doi: 10.1016/j.compfluid.2011.02.014.
- [4] Xiangbin Chen, Tianyang Han, Jieyun Pan, Daniel Fuster, and Stéphane Zaleski. Volume-conserving method for dynamic contact line on complex surfaces. *Physics of Fluids*, 37(2):023392, 2025. doi: 10.1063/5.0254089.
- [5] G. Colicchio, M. Greco, and O. M. Faltinsen. A BEM–level set domain-decomposition strategy for non-linear and fragmented interfacial flows. *International Journal for Numerical Methods in Engineering*, 67(10):1385–1419, 2006. doi: 10.1002/nme.1680.
- [6] J. W. Dold and D. H. Peregrine. An efficient boundary integral method for steep unsteady water waves. In K. W. Morton and M. J. Baines, editors, *Numerical Methods for Fluid Dynamics II*, pages 671–679. Oxford University Press, Oxford, 1986.
- [7] Christian Gorges, Fabien Evrard, Berend van Wachem, and Fabian Denner. Reducing volume and shape errors in front tracking by divergence-preserving velocity interpolation and parabolic fit vertex positioning. *Journal of Computational Physics*, 457:111072, 2022. doi: 10.1016/j.jcp.2022.111072.
- [8] S. T. Grilli, I. A. Svendsen, and R. Subramanya. Breaking criterion and characteristics for solitary waves on slopes. *Journal of Waterway, Port, Coastal, and Ocean Engineering*, 123(3):102–112, 1997. doi: 10.1061/(ASCE)0733-950X(1997)123:3(102).
- [9] C. W. Hirt and B. D. Nichols. Volume of fluid (VOF) method for the dynamics of free boundaries. *Journal of Computational Physics*, 39(1):201–225, 1981. doi: 10.1016/0021-9991(81)90145-5.

- [10] Chong-Sen Huang, Tian-Yang Han, Jie Zhang, and Ming-Jiu Ni. A 2d sharp and conservative VOF method for modeling the contact line dynamics with hysteresis on complex boundary. *Journal of Computational Physics*, 533:113975, 2025. doi: 10.1016/j.jcp.2025.113975.
- [11] Chong-Sen Huang, Tian-Yang Han, Jie Zhang, and Ming-Jiu Ni. A 3d sharp and conservative VOF method for modeling the contact line dynamics with hysteresis on complex boundaries. *Journal of Computational Physics*, 566:115228, 2026. doi: 10.1016/j.jcp.2026.115228.
- [12] Taiga Kanehira, Peter K. Stansby, Benedict D. Rogers, Mark McAllister, T. S. van den Bremer, and Samuel Draycott. Dynamics of jet formation and collapse for axisymmetric surface gravity waves: coupled 3-d potential flow and SPH simulations. *Journal of Fluid Mechanics*, 1041:A17, 2026. doi: 10.1017/jfm.2026.11914.
- [13] Anatoliy Khait and Zhihua Ma. On an eddy viscosity model for energetic deep-water surface gravity wave breaking. *Journal of Fluid Mechanics*, 929:A29, 2021. doi: 10.1017/jfm.2021.863.
- [14] Sang Ho Kim, Masaru Yamashiro, and Akinori Yoshida. A simple two-way coupling method of BEM and VOF model for random wave calculations. *Coastal Engineering*, 57(11–12):1018–1028, 2010. doi: 10.1016/j.coastaleng.2010.06.006.
- [15] Christophe Lachaume, Benjamin Biaisser, Stéphane T. Grilli, Philippe Fraunié, and Stéphane Guignard. Modeling of breaking and post-breaking waves on slopes by coupling of BEM and VOF methods. In *Proceedings of the 13th International Offshore and Polar Engineering Conference*, pages 353–359, Honolulu, Hawaii, 2003. International Society of Offshore and Polar Engineers. URL https://digitalcommons.uri.edu/oce_facpubs/177/.
- [16] M. S. Longuet-Higgins and E. D. Cokelet. The deformation of steep surface waves on water. i. a numerical method of computation. *Proceedings of the Royal Society of London A*, 350:1–26, 1976. doi: 10.1098/rspa.1976.0092.
- [17] Stéphane Popinet. Gerris: a tree-based adaptive solver for the incompressible Euler equations in complex geometries. *Journal of Computational Physics*, 190(2):572–600, 2003. doi: 10.1016/S0021-9991(03)00298-5.
- [18] Stéphane Popinet. An accurate adaptive solver for surface-tension-driven interfacial flows. *Journal of Computational Physics*, 228(16):5838–5866, 2009. doi: 10.1016/j.jcp.2009.04.042.
- [19] Alan Riquier and Emmanuel Dormy. Numerical study of a viscous breaking water wave and the limit of vanishing viscosity. *Journal of Fluid Mechanics*, 984:R5, 2024. doi: 10.1017/jfm.2024.208.
- [20] Shuyu Sun and Mary F. Wheeler. Projections of velocity data for the compatibility with transport. *Computer Methods in Applied Mechanics and Engineering*, 195(7–8):653–673, 2006. doi: 10.1016/j.cma.2005.02.011.
- [21] Mathilde Tavares, Christophe Josserand, Alexandre Limare, José-María Lopez-Herrera, and Stéphane Popinet. A coupled VOF/embedded boundary method to model two-phase flows on arbitrary solid surfaces. *Computers & Fluids*, 278:106317, 2024. doi: 10.1016/j.compfluid.2024.106317.